

Basic Circuits

T.U.B.U.R.A.N. Learning Module

Voltage, Current, and Resistance

Voltage (V) is the electrical pressure that pushes current through a circuit, measured in volts.

Current (I) is the flow of electric charge, measured in amperes (amps). Resistance (R) opposes the flow of current, measured in ohms.

Ohm's Law

Ohm's Law describes the relationship between these three quantities: $V = I \times R$. This means voltage equals current multiplied by resistance. If you know any two values, you can calculate the third.

Building a Simple Circuit

A basic series circuit needs three things: a power source (like a battery), a load (like an LED or motor), and a conductive path (wires) connecting them in a closed loop. Current only flows when the loop is complete.

The Role of a Resistor

An LED alone connected directly to a battery can burn out because too much current flows through it. A resistor limits the amount of current, protecting the LED while still allowing enough current for it to light up properly.

Series vs Parallel Circuits

In a series circuit, components are connected end-to-end, so current has only one path to follow.

In a parallel circuit, components are connected across common points, giving current multiple paths. Series circuits share voltage; parallel circuits share current.